

Mass transfer to small particles < 0.6 mm	$k'_L = \frac{2D_{AB}}{D_p} + 0.31N_{Sc}^{-2/3} \left(\frac{\Delta\rho\mu_c g}{\rho_c^2} \right)^{1/3}$	(21.3-45)
Mass transfer to large gas bubbles > 2.5 mm	$k'_L = 0.42N_{Sc}^{-0.5} \left(\frac{\Delta\rho\mu_c g}{\rho_c^2} \right)^{1/3}$	(21.4-1)
Mass transfer to particles in highly turbulent mixers	$k'_L N_{Sc}^{2/3} = 0.13 \left(\frac{P/V}{\rho_c^2} \mu_c \right)^{1/4}$	(21.4-3)
		(21.4-4)

Models for Mass-Transfer Coefficients

a. *Film mass-transfer theory.*

$$J_A^* = k'_c(c_{A1} - c_{A2}) = \frac{D_{AB}}{\delta_f}(c_{A1} - c_{A2}) \quad (21.5-17)$$

$$k'_c = \frac{D_{AB}}{\delta_f} \quad (21.5-18)$$

b. *Penetration theory.*

$$k'_c = \sqrt{\frac{4D_{AB}}{\pi t_L}} \quad (21.5-19)$$

$$k'_c = \sqrt{D_{AB}s} \quad (21.5-20)$$

c. *Boundary-layer theory.*

$$\frac{k'_c L}{D_{AB}} = N_{Sh} = 0.664 N_{Re,L}^{1/2} N_{Sc}^{1/3} \quad (21.5-10)$$

PROBLEMS

21.1-1. Flux and Conversion of Mass-Transfer Coefficient. A value of k_G was experimentally determined to be $1.08 \text{ lb mol/h} \cdot \text{ft}^2 \cdot \text{atm}$ for A diffusing through stagnant B . For the same flow and concentrations, it is desired to predict k'_G and the flux of A for equimolar counterdiffusion. The partial pressures are $p_{A1} = 0.20 \text{ atm}$, $p_{A2} = 0.05 \text{ atm}$, and $P = 1.0 \text{ atm}$ abs total. Use English and SI units.

Ans. $k'_G = 0.943 \text{ lb mol/h} \cdot \text{ft}^2 \cdot \text{atm}$, $1.262 \times 10^{-8} \text{ kg mol/s} \cdot \text{m}^2 \cdot \text{Pa}$,
 $N_A = 0.1414 \text{ lb mol A/h} \cdot \text{ft}^2$, $1.918 \times 10^{-4} \text{ kg mol/s} \cdot \text{m}^2$

21.1-2. Conversion of Mass-Transfer Coefficients. Prove or show the following relationships, starting with the flux equations:

- Convert k'_c to k_y and k_G .
- Convert k_L to k_x and k'_x .
- Convert k_G to k_y and k_c .

21.1-3. Absorption of H_2S by Water. In a wetted-wall tower an air- H_2S mixture is flowing by a film of water that is flowing as a thin film down a vertical plate. The H_2S is being absorbed from the air to the water at a total pressure of 1.50 atm abs and $30^\circ C$. A value for k'_c of 9.567×10^{-4} m/s has been predicted for the gas-phase mass-transfer coefficient. At a given point, the mole fraction of H_2S in the liquid at the liquid-gas interface is $2.0(10^{-3})$ and p_A of H_2S in the gas is 0.05 atm. The Henry's law equilibrium relation is $p_A(\text{atm}) = 609x_A$ (mole fraction in liquid). Calculate the rate of absorption of H_2S . (Hint: Call point 1 the interface and point 2 the gas phase. Then, calculate p_{A1} from Henry's law and the given x_A . The value of p_{A2} is 0.05 atm.)

Ans. $N_A = -1.485 \times 10^{-6}$ kg mol/s $\cdot$ m²

21.1-4. Effect of High Flux on Mass-Transfer Coefficients. Using the data from Example 21.1-2, also calculate the flux ratios k'_y/k'_x for $x_{A1} = 0.20$ and $x_{A1} = 0.01$. Tabulate these ratios for the three values of x_{A1} and plot the ratio versus x_{A1} .

21.2-1. Dimensional Analysis in Mass Transfer. A fluid is flowing in a vertical pipe and mass transfer is occurring from the pipe wall to the fluid. Relate the convective mass-transfer coefficient k'_c to the variables D , ρ , μ , v , D_{AB} , g , and $\Delta\rho$, where D is pipe diameter, L is pipe length, and $\Delta\rho$ is the density difference.

$$\text{Ans. } \frac{k'_c D}{D_{AB}} = f \left(\frac{gL^3 \rho \Delta\rho}{\mu^2}, \frac{Dv\rho}{\mu}, \frac{\mu}{\rho D_{AB}} \right)$$

21.3-1. Mass Transfer from a Flat Plate to a Liquid. Using the data and physical properties of Example 21.3-2, calculate the flux for a water velocity of 0.152 m/s and a plate length of $L = 0.137$ m. Do not assume that $x_{BM} = 1.0$ but actually calculate its value.

21.3-2. Mass Transfer from a Pipe Wall. Pure water at $26.1^\circ C$ is flowing at a velocity of 0.0305 m/s in a tube having an inside diameter of 6.35 mm. The tube is 1.829 m long, with the last 1.22 m having the walls coated with benzoic acid. Assuming that the velocity profile is fully developed, calculate the average concentration of benzoic acid at the outlet. Use the physical property data from Example 21.3-2. [Hint: First, calculate the Reynolds number $Dv\rho/\mu$. Then, calculate $N_{Re}N_{Sc}(D/L)(\pi/4)$, which is the same as $W/D_{AB}\rho L$.]

Ans. $(c_A - c_{A0})/(c_{A1} - c_{A0}) = 0.0744$, $c_A = 2.193 \times 10^{-3}$ kg mol/m³

21.3-3. Mass-Transfer Coefficient for Various Geometries. It is desired to estimate the mass-transfer coefficient k_G in kg mol/s $\cdot$ m² $\cdot$ Pa for water vapor in air at 338.6 K and 101.32 kPa flowing in a large duct past solids of different geometries. The velocity in the duct is 3.66 m/s. The water vapor concentration in the air is small, so the physical properties of air can be used. Water vapor is being transferred to the solids. Do this for the following geometries:

- A single 25.4-mm-diameter sphere.
- A packed bed of 25.4-mm spheres with $\varepsilon = 0.35$.

Ans. (a) $k_G = 1.984 \times 10^{-8}$ kg mol/s $\cdot$ m² $\cdot$ Pa (1.482 lb mol/h $\cdot$ ft² atm)

21.3-4. Mass Transfer to Definite Shapes. Estimate the value of the mass-transfer coefficient in a stream of air at 325.6 K flowing in a duct past the following shapes made of solid naphthalene. The velocity of the air is 1.524 m/s at 325.6 K and 202.6 kPa. The D_{AB} of naphthalene in air is 5.16×10^{-6} m²/s at 273 K and 101.3 kPa.

- For air flowing parallel to a flat plate 0.152 m in length
- For air flowing past a single sphere 12.7 mm in diameter

21.3-5. Mass Transfer to Packed Bed and Driving Force. Pure water at $26.1^\circ C$ is flowing at a rate of 0.0701 ft³/h through a packed bed of 0.251-in. benzoic-acid spheres having a total surface area of 0.129 ft². The solubility of benzoic acid in water is 0.00184 lb

mol benzoic acid/ft³ solution. The outlet concentration c_{A2} is 1.80×10^{-4} lb mol/ft³. Calculate the mass-transfer coefficient k_c .

- 21.3-6. Mass Transfer in Liquid Metals.** Mercury at 26.5°C is flowing through a packed bed of lead spheres having a diameter of 2.096 mm with a void fraction of 0.499. The superficial velocity is 0.02198 m/s. The solubility of lead in mercury is 1.721 wt %, the Schmidt number is 124.1, the viscosity of the solution is 1.577×10^{-3} Pa · s, and the density is 13 530 kg/m³.

- (a) Predict the value of J_D . Use Eq. (21.3-38) if applicable. Compare with the experimental value of $J_D = 0.076$ (D2).
(b) Predict the value of k_c for the case of A diffusing through nondiffusing B.

Ans. (a) $J_D = 0.0784$; (b) $k_c = 6.986 \times 10^{-5}$ m/s

- 21.3-7. Mass Transfer from a Pipe and Log Mean Driving Force.** Use the same physical conditions as in Problem 21.3-2, but the velocity in the pipe is now 3.05 m/s. Do as follows:

- (a) Predict the mass-transfer coefficient k'_c . (Is this turbulent flow?)
(b) Calculate the average benzoic acid concentration at the outlet. [Note: In this case, Eqs. (21.3-42) and (21.3-43) must be used with the log mean driving force, where A is the surface area of the pipe.]
(c) Calculate the total kg mol of benzoic acid dissolved per second.

- 21.3-8. Derivation of Relation between J_D and N_{Sh} .** Equation (21.3-3) defines the Sherwood number and Eq. (21.3-5) defines the J_D factor. Derive the relation between N_{Sh} and J_D in terms of N_{Re} and N_{Sc} .

Ans. $N_{Sh} = J_D N_{Re} N_{Sc}^{1/3}$

- 21.3-9. Driving Force to Use in Mass Transfer.** Derive Eq. (21.3-42) for the log mean driving force to use for a fluid flowing in a packed bed or in a tube. (Hint: Start by making a mass balance and a diffusion-rate balance over a differential area dA as follows:

$$N_A dA = k_c (c_{A1} - c_A) dA = V dc_A$$

where $V = \text{m}^3/\text{s}$ flow rate. Assume dilute solutions.)

- 21.4-1. Maximum Oxygen Uptake of a Microorganism.** Calculate the maximum possible rate of oxygen uptake at 37°C of a microorganism having a diameter of $\frac{4}{3} \mu\text{m}$ suspended in an agitated aqueous solution. It is assumed that the surrounding liquid is saturated with O₂ from air at 1 atm abs pressure. It will be assumed that the microorganism can utilize the oxygen much faster than it can diffuse to it. The microorganism has a density very close to that of water. Use physical-property data from Example 21.4-1. (Hint: Since the oxygen is consumed faster than it is supplied, the concentration c_{A2} at the surface is zero. The concentration c_{A1} in the solution is at saturation.)

Ans. $k_c = 9.75 \times 10^{-3}$ m/s, $N_A = 2.20 \times 10^{-6}$ kg mol O₂/s · m²

- 21.4-2. Mass Transfer of O₂ in Fermentation Process.** A total of 5.0 g of wet microorganisms having a density of 1100 kg/m³ and a diameter of 0.667 μm are added to 0.100 L of aqueous solution at 37°C in a shaker flask for a fermentation. Air can enter through a porous stopper. Use physical property data from Example 21.4-1.
(a) Calculate the maximum rate possible for mass transfer of oxygen in kg mol O₂/s to the surface of the microorganism, assuming that the solution is saturated with air at 101.32 kPa abs pressure.

HW #3

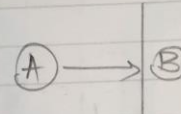
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ANDRIES - BRAGGQ1.1-1 FLUX AND CONVERSION OF MASS-TRANSFER COEFFICIENT

$$k_g = 1.08 \text{ lb} \cdot \text{mol} / \text{h} \cdot \text{ft}^2$$

$$p_{A1} = 0.20 \text{ atm}$$

$$p_{A2} = 0.05 \text{ atm}$$

$$P = 1.0 \text{ atm}$$



Given the case of
equimolar counterdiffusion,
we use:

(A) diffusing through
stagnant (B)

$$k'_g P \equiv k_g P_{BM}, \text{ where}$$

$$P_{BM} = \frac{p_{A1} - p_{A2}}{\ln(P - p_{A2}) / (P - p_{A1})}$$

$$\Rightarrow k'_g = \frac{P_{BM} k_g}{P} = \frac{k_g (p_{A1} - p_{A2})}{P \ln \frac{P - p_{A2}}{P - p_{A1}}}$$

$$\Rightarrow k'_g = \frac{(1.08 \text{ lb} \cdot \text{mol} / \text{h} \cdot \text{ft}^2) (0.20 \text{ atm} - 0.05 \text{ atm})}{(1.0 \text{ atm}) \ln \frac{1 - 0.05}{1 - 0.20}}$$

$$= \frac{(1.08)(0.15)}{\ln(0.95/0.8)} = \frac{0.162}{\ln(1.1875)} = 0.943 \text{ lb} \cdot \text{mol} / \text{h} \cdot \text{ft}^2 \cdot \text{atm}$$

$$k'_g (\text{SI units}) = \left(0.943 \frac{\text{lb mol}}{\text{h ft}^2 \text{ atm}} \right) \cdot \left(\frac{0.454 \text{ kg}}{1 \text{ lb}} \right) \cdot \left(\frac{1 \text{ h}}{3600 \text{ s}} \right)$$

$$= \left(\frac{1 \text{ ft}^2}{0.0929 \text{ m}^2} \right) \cdot \left(\frac{1 \text{ atm}}{101325 \text{ Pa}} \right)$$

$$\Rightarrow k'_g (\text{SI}) = 1.262 \times 10^{-8} \frac{\text{kg mol}}{\text{s m}^2 \text{ Pa}}$$

For flux of (A), we use:

$$N_A = k'_g (p_{A1} - p_{A2}) =$$

$$= \left(0.943 \frac{\text{lb mol}}{\text{h ft}^2 \text{ atm}} \right) (0.20 \text{ atm} - 0.05 \text{ atm}) =$$

$$= (0.943) (0.15) = 0.14145 \frac{\text{lb mol}}{\text{h ft}^2}$$

$$\approx 0.1415 \frac{\text{lb mol}}{\text{h ft}^2}$$

Thus:

$k'_g (\text{English})$	$0.943 \text{ lb mol/h ft}^2 \text{ atm}$
$k'_g (\text{SI})$	$1.262 \times 10^{-8} \text{ kg mol/s m}^2 \text{ Pa}$
N_A	$0.1415 \text{ lb mol/h ft}^2$

2.1.1-3 ABSORPTION OF H_2S BY WATER

$$T = 30^\circ C = 303.15 K$$

$$P_T = 1.50 \text{ atm}$$

$$P_{A2} = 0.05 \text{ atm}$$

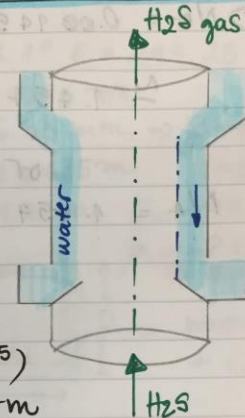
$$x_A = 2.0 \times 10^{-5}$$

$$k'_c = 9.567 \times 10^{-4} \text{ m/s}$$

Henry's Law:

$$P_{A1} = 609 x_A$$

$$\Rightarrow P_{A1} = (609)(2.0 \times 10^{-5}) = 0.01218 \text{ atm}$$



We want to find: $N_A = k'_c (C_{A1} - C_{A2}) (2.1-5)$
 where $C = \frac{P}{RT} \Rightarrow C_{A1} = \frac{P_{A1}}{RT}$

$$C_{A2} = \frac{P_{A2}}{RT}$$

$$\Rightarrow N_A = \frac{k'_c}{RT} (P_{A1} - P_{A2})$$

$$= \frac{9.567 \times 10^{-4} \text{ m/s}}{(8.314 \frac{\text{J}}{\text{mol} \cdot \text{K}})(303.15 \text{ K})} (0.01218 \text{ atm} - 0.05 \text{ atm})$$

$$= \frac{(9.567 \times 10^{-4}) \text{ mol}}{(2520.389) \text{ m}^2 \cdot \text{Pa} \cdot \text{s}} (-0.0378 \text{ atm}) \cdot \frac{101325 \text{ Pa}}{1 \text{ atm}}$$

5.1-3 ABSORPTION OF H₂ BY WATER

$$\Rightarrow N_A = 0.00145384 \text{ mol/m}^2 \cdot \text{s}$$

$$\approx 1.454 \times 10^{-3} \text{ mol/m}^2 \cdot \text{s}$$

or

$$N_A = 1.454 \times 10^{-6} \text{ kmol/m}^2 \cdot \text{s}$$

21.3-2 Mass Transfer From a Pipe Wall.

(A) Benzoic Acid $T = 26.1^\circ \text{C} \equiv 299.25 \text{ K}$
 (B) water $v = 0.0305 \text{ m/s}$
 $C_{Ai} \equiv \text{solubility} = 0.02948 \frac{\text{kg}}{\text{m}^3} \text{ C}_7\text{H}_6\text{O}_2$
 $D_{AB} \equiv \text{diffusivity} = 1.2 \times 10^{-9} \text{ m}^2/\text{s} \text{ C}_7\text{H}_6\text{O}_2$
 $D = 6.35 \text{ mm} \Rightarrow R = 3.175 \text{ mm} = 0.003175 \text{ m}$
 $L = 1.22 \text{ m}$
 $C_w = 997 \text{ kg/m}^3$
 $\mu_w = 8.9 \times 10^{-4} \text{ Pa}\cdot\text{s}$

C_{A0} inlet conc.
 water
 C_{A0} inlet conc.
 water
 C_{Ai} outlet conc.
 water + $\text{C}_7\text{H}_6\text{O}_2$

Solution : Calculating Reynold's No. :

$$\begin{aligned}
 Re &= Dv\rho/\mu = (0.00635 \text{ m})(0.0305 \text{ m/s})(997 \text{ kg/m}^3) / (8.9 \times 10^{-4} \text{ Pa}\cdot\text{s}) \\
 &= \frac{(0.00635)(0.0305)(997)}{(8.9 \times 10^{-4})} \frac{\text{m} \cdot \text{m/s} \cdot \text{kg/m}^3}{(\text{kg/m}\cdot\text{s}) (\text{s})}
 \end{aligned}$$

$= 216.96 \Rightarrow$ flow is laminar.

$2100 > Re \Rightarrow$ laminar

$2100 > Re > 4000 \Rightarrow$ transitive

$Re > 4000 \Rightarrow$ turbulent.

Now we calculate

$$N_{Re} N_{Sc} \left(\frac{D}{L} \right) \left(\frac{\pi}{4} \right) = \left(\frac{W}{D_{AB} \cdot \rho \cdot L} \right)$$

where

$$N_{Sc} = \frac{\mu}{\rho D_{AB}} = \frac{(0.8718 \times 10^{-3}) (\text{Pa} \cdot \text{s})}{(996.7 \frac{\text{kg}}{\text{m}^3}) (1.21 \times 10^{-9} \frac{\text{m}^2}{\text{s}})}$$

$$\Rightarrow N_{Sc} = 722.88 \text{ . Then}$$

$$N_{Re} N_{Sc} \left(\frac{\pi D}{4L} \right) =$$

$$= (216.96)(722.88) \left(\frac{\pi \cdot 0.00635 \text{ m}}{4 (1.22 \text{ m})} \right)$$

$$\Rightarrow 641.136 = \left(\frac{W}{D_{AB} \rho L} \right)$$

knowing that for liquids the following eqn applies, we use:

$$\frac{C_A - C_{A0}}{C_{Ai} - C_{A0}} = 5.5 \left(\frac{W}{D_{AB} \rho L} \right)^{-2/3} \text{ We want to}$$

solve for C_A , thus we rearrange:

$$C_A - C_{A0} = 5.5 \left(\frac{w}{D_{AB} \rho L} \right)^{-2/3} (C_{Ai} - C_{A0})$$

since there is no C_{A0} at the inlet

$$\Rightarrow C_A = 5.5 \left(\frac{w}{D_{AB} \rho L} \right)^{-2/3} (C_{Ai} - C_{A0}) + C_{A0}$$

$$\Rightarrow C_A = 5.5 (641.136)^{-2/3} (0.02948 \text{ kg mol/m}^3)$$

$$= 0.00218064 \approx 2.181 \times 10^{-3} \text{ kg mol/m}^3$$

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21.3-5 Mass Transfer to Packed Bed and Driving Force

$$T = 26.1^\circ \text{C}$$

$$V = 0.0401 \text{ ft}^3/\text{hr}$$

$$D = 0.25 \text{ in}$$

$$A = 0.129 \text{ m}^2$$

$$C_A^S = 0.00184 \text{ kmol/m}^3$$

$$C_{A2} = 1.80 \times 10^{-4} \text{ lb mol/ft}^3$$

We know that molar flux for (A) diffusing through a stagnant, nondiffusing (B) ($\equiv N_B = 0$) is given by:

$$N_A = -k_c (C_{A1} - C_{A2})$$

where we consider $C_{A1} = C_A^S$, the solubility of benzene acid in water.

At the same time, according to the 21.3E-3 section in the textbook, the material balance equation on the bulk stream is:

$$N_A A = V (C_{A2} - C_{A1})$$

Thus we have 2 equations and one unknown as such:

$$\begin{cases} N_A = \underline{k_c} (C_{A1}^s - C_{A2}) \\ N_A A = V (C_{A2} - C_{A1}) \end{cases}$$

=> Solving for k_c :

$$k_c (C_{A1}^s - C_{A2}) \cdot A = V (C_{A2} - C_{A1})$$

$$k_c = \frac{V (C_{A2} - C_{A1})}{A (C_{A1}^s - C_{A2})} = \frac{(0.0701 \text{ ft}^3/\text{hr}) (1.80 \times 10^{-4} \text{ lb mol/ft}^3)}{(0.129 \text{ m}^2) (1.80 \times 10^{-4} \text{ lb mol/ft}^3) - (0.00184 \text{ kmol/m}^3)}$$

$$\Rightarrow k_c = \frac{\left(\frac{0.0701 \text{ ft}^3}{3600 \text{ s}} \right) \left(\frac{1.80 \times 10^{-4} \text{ lb mol}}{\text{ft}^3} \right)}{\left(\frac{0.129}{\text{m}^2} \right) \left(\frac{1.80 \times 10^{-4} \text{ lb mol}}{\text{ft}^3} \right) - \left(\frac{0.00184 \text{ kmol}}{\text{ft}^3} \right) \left(\frac{0.0624 \text{ lb mol}}{\text{ft}^3} \right)}$$

$$\Rightarrow k_c = \frac{(1.9472 \times 10^{-5} \text{ ft}^3/\text{s}) (1.80 \times 10^{-4} \text{ lbmol}/\text{ft}^3)}{(0.129) \left(\frac{1.80 \times 10^{-4} \text{ lbmol}/\text{ft}^3}{\text{m}^2} - \frac{1.148 \times 10^{-4} \text{ lbmol}/\text{ft}^3}{\text{m}^2} \right)}$$

$$\Rightarrow k_c = 0.05 \text{ m/s} \rightarrow \text{I think this should be ft/h to work}$$

21.3-6 Mass Transfer in Liquid Metals.

(A) = Lead (Pb)

(B) = Mercury (Hg)

$$T = 26.5^\circ \text{C}$$

$$D = 0.002096 \text{ m}$$

$$\varepsilon = 0.999$$

$$\nu' = 0.02198 \text{ m}^2/\text{s}$$

$$\mu = 1.577 \times 10^{-3} \text{ Pa} \cdot \text{s}$$

$$\rho = 13530 \text{ kg}/\text{m}^3$$

$$N_{sc} = 124.1$$

$$N_{Re} = \frac{D \nu' \rho}{\mu} \Rightarrow$$

$$(a) \Rightarrow N_{Re} = \frac{(0.002096 \text{ m})(0.02198 \text{ m}^2/\text{s})(13530 \text{ kg}/\text{m}^3)}{(1.577 \times 10^{-3} \text{ kg}/\text{m} \cdot \text{s})(\text{s})}$$

$$\Rightarrow N_{Re} = \frac{0.6233}{1.577 \times 10^{-3}} = 395.262 \text{ (Laminar Flow)}$$

Having computed a value for Reynold's number, we can predict a J_D value.

Eq. 21.3-38:

$$J_D = \frac{0.250}{E} N_{Re}^{-0.31}$$

$$= \frac{0.250}{0.499} (395.262)^{-0.31}$$

$$\Rightarrow J_D = 0.07849$$

$$J_{D, \text{experimental}} = 0.076$$

} $\Rightarrow$ analytical value
close enough to
experimental J_D .

$$\text{Percent error} = \frac{(0.076) - (0.0785)}{0.0785} \times 100\% = -3.185\%$$

3.185% deviation from
experimental value

(b) (A) diffusing through non-diffusing species (B) ; $\begin{cases} M_A = 207.19 \text{ moles} \\ M_B = 200.59 \text{ moles} \end{cases}$

We assume a total mass of 100 kg for the saturated solution $\Rightarrow$

$$\Rightarrow m_{\text{sol}} = 100 \text{ kg} \quad \text{wt \% A} = 1.721 \quad \Rightarrow m_A = \frac{1.721}{100} \times 100 \text{ kg}$$

$$\Rightarrow m_B = 100 - 1.721 \Rightarrow \begin{cases} m_A = 1.721 \text{ kg} \\ m_B = 98.279 \text{ kg} \end{cases}$$

Then we can calculate the mole fraction of (A), (B).

$$x_{A2} = \frac{\frac{m_A}{M_A}}{\frac{m_A}{M_A} + \frac{m_B}{M_B}} = \frac{\frac{1.721}{207.19}}{\frac{1.721}{207.19} + \frac{98.279}{200.59}} \Rightarrow$$

$$\Rightarrow x_{A1} = \frac{0.008306}{0.008306 + 0.48995} = 0.01667$$

But $x_{A2} = 0$, since lead does not leave the system.

$$x_{B1} = x_{\text{total}} - x_{A1}$$

$$= 1.00 - 0.01667$$

$$\Rightarrow x_{B1} = 0.9833$$

$$\text{And } x_{B2} = 1.00 - 0$$

$= 1.00$, since only mercury is the species flowing outside of study system.

Then we can calculate the average molar fraction of species (B):

$$x_{BM} = \frac{x_{B1} + x_{B2}}{2}$$

$$= \frac{(0.9833 + 1.000)}{2} = 1.9833$$

$$\Rightarrow x_{BM} = 0.9917$$

We want to predict a value for k_c .

$$(2.1.1-10) \quad \frac{k'_c}{N_A} (C_{A1} - C_{A2}) = (k_c) (C_{A1} - C_{A2}) \left| \div (C_{A1} - C_{A2}) \right.$$

$$N_A = x_{BM}$$

→ unknown.

$$\Rightarrow \frac{k'_c}{x_{BM}} = k_c$$

→ known 12

From (21.3-5) :

$$f_D = \frac{(k_c')}{v'} (N_{Sc})^{2/3} \text{ . Solving for } k_c' =$$

$$\Rightarrow k_c' = \frac{f_D v'}{(N_{Sc})^{2/3}} = \frac{(0.07849)(0.02198 \text{ m/s})}{(124.1)^{2/3}}$$

$$\Rightarrow k_c' = \frac{0.001725 \text{ m/s}}{24.8799} = 6.9342 \times 10^{-5} \text{ m/s}$$

Then we can finally calculate k_c :

$$k_c = \frac{k_c'}{\alpha_{BM}} = \frac{6.934 \times 10^{-5} \text{ m/s}}{0.9917}$$

$$\Rightarrow k_c = 6.9922 \times 10^{-5} \text{ m/s. ! Nice!}$$

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